Replication Assessment: Aid Under Fire

Applied Causal Inference, Week 6 Day 4

Harris Social Impact Fellowship · Summer 2026

How today runs

Today is yours. Four stages, roughly two hours of team work, 9:30–11:45. Austin and Prabidhik circle the room; flag either of us at any point. Work the stages in sequence; each feeds a section of your memo, and the starter script (`hsif_week6_day4_aidunderfire_helper.R`) is organized to match.

Target: Crost, Felter & Johnston (2014), "Aid under Fire: Development Projects and Civil Conflict," *AER* 104(6): 1833–56. The replication package (openICPSR 112831) is on Canvas: analysis data, code, and the kit README. Fill the starter script's variable map from the package codebook before estimating anything.

Deliverable: a team memo, roughly 1,200 words plus figure arrays, two to three pages total. Drafted by 11:45; the finalized memo is due by the end of the day.

The memo

Four sections, mapping to the four stages:

1. **Setting and inference problem** (∼250 words): the design, and what the reconstructed ranking does to it
2. **Main results, modernized** (∼350 words + figure array 1): the headline effect under current standards
3. **Heterogeneity under scrutiny** (∼300 words): the timing and actor-level claims, with your validity verdicts
4. **The complement** (∼300 words + figure array 2): the DiD event study and the RD event study, and what survives

Figure arrays: small multiples, labeled, readable in grayscale. Write for the referee you have practiced being all week: assumptions named, verdicts committed, figures carrying the argument.

Stage 1 (9:30–9:55): context and the ranking problem

Read the paper's design section and the package codebook. No estimation yet.

The program ranked municipalities by poverty within each province; the poorest quartile became eligible. The files contain a ranking **reconstructed from published poverty estimates**: the program's internal list, and any discretionary adjustments to it, are not available to you or to the authors.

Work the consequences, in writing:

- Misclassification near the cutoff: some municipalities sit on the wrong side of your reconstructed line. What does that do to the eligibility jump you can estimate?
- Which validity checks become indirect? What is your density test actually testing?
- Does measurement error in the running variable push your estimates toward or away from zero, and why?
- Where does misclassification concentrate, and which specification in the toolkit targets exactly that?

Memo section 1: the design in three sentences, then the inference problem in five.

Stage 2 (9:55–10:40): main results, modernized

Replicate, then push. Starter script Block 2.

1. **Replicate**: the main casualty RD at the paper's specification (local linear, triangular kernel, $h = 6$ ranks). Confirm you hit the published estimate before changing anything; a miss is a variable-map error, not a finding
2. **RD plot**: binned means against the centered rank, fits on each side. The jump should be visible before any table confirms it
3. **Modernize**: `rdrobust` at the MSE-optimal bandwidth: Conventional and Robust rows, then $h/2$ and $2h$
4. **Donut**: drop ranks within $r \in \{1, 2\}$ of the cutoff and re-estimate: the direct answer to Stage 1's misclassification concern

Memo section 2 + figure array 1: the RD plot and a specification chart (six estimates with confidence intervals on one axis: paper's spec, MSE-optimal, $h/2$, $2h$, donut 1, donut 2). One paragraph: does the headline survive, and on which rows does the answer rest?

Break as a team when you need one, ten minutes, before Stage 3.

Stage 3 (10:40–11:10): timing and actors under scrutiny

The paper's mechanism: casualties rise in the program-preparation stage, driven by insurgent-initiated incidents. Sabotage before benefits arrive. Starter script Block 3.

- **Actor-level results**: casualties split by initiator are separate outcomes, each its own RD. Re-estimate both with robust rows. Then interrogate the measurement: initiator coding comes from military records; could coding practices differ across the eligibility line? Propose one check and run it if the files allow

- **Timing results**: the stage windows (preparation vs. implementation) are defined by the *treated* municipalities' program calendar, and rollout was staggered. What calendar do barely ineligible municipalities get assigned, and is it exogenous? If rollout timing tracked local conditions, which of Tuesday's problems is now inside the RD?

Memo section 3: two verdicts, one per claim: theoretically compelling; econometrically valid, or not, and why.

Stage 4 (11:10–11:45): the complement, and a better clock

The files are a municipality-by-period panel. Use it. Starter script Block 4.

1. **The DiD complement**: event study around program start, eligible vs. ineligible. Tuesday's checklist in full: reference period, window, summed endpoint bins, pre-trends, the honest-sensitivity sentence. Does the RD result survive under a different assumption?
2. **The RD event study**: instead of the authors' three coarse periods, estimate the RD *period by period*: for each event time k, the jump in casualties at the cutoff, plotted as a path with intervals. Leads are built-in placebos: jumps before the program should be zero. The path shows onset, peak, and fade that three periods compress. One estimate per period costs power; read the intervals honestly

Memo section 4 + figure array 2: both event-study figures side by side. Closing paragraph: what survives everything you threw at it today?

The toolkit, on one page

- **Robust inference**: `rdrobust` at the MSE-optimal h; report the Robust row; $h/2$ and $2h$ alongside
- **Donut**: drop ranks within r of the cutoff; re-estimate; report sensitivity to r
- **Fuzzy logic**: sharp estimation on eligibility is an ITT at the cutoff; the Wald correction (sharp ÷ first-stage jump in receipt) prices the program, for cutoff compliers
- **Density for ranks**: uniform by construction; test the underlying score
- **Event studies**: reference period stated, window justified, endpoint bins summed, pre-trends plotted and jointly tested

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By 11:45: all four stages estimated, memo drafted, figures placed. Submit the finalized memo by the end of the day.